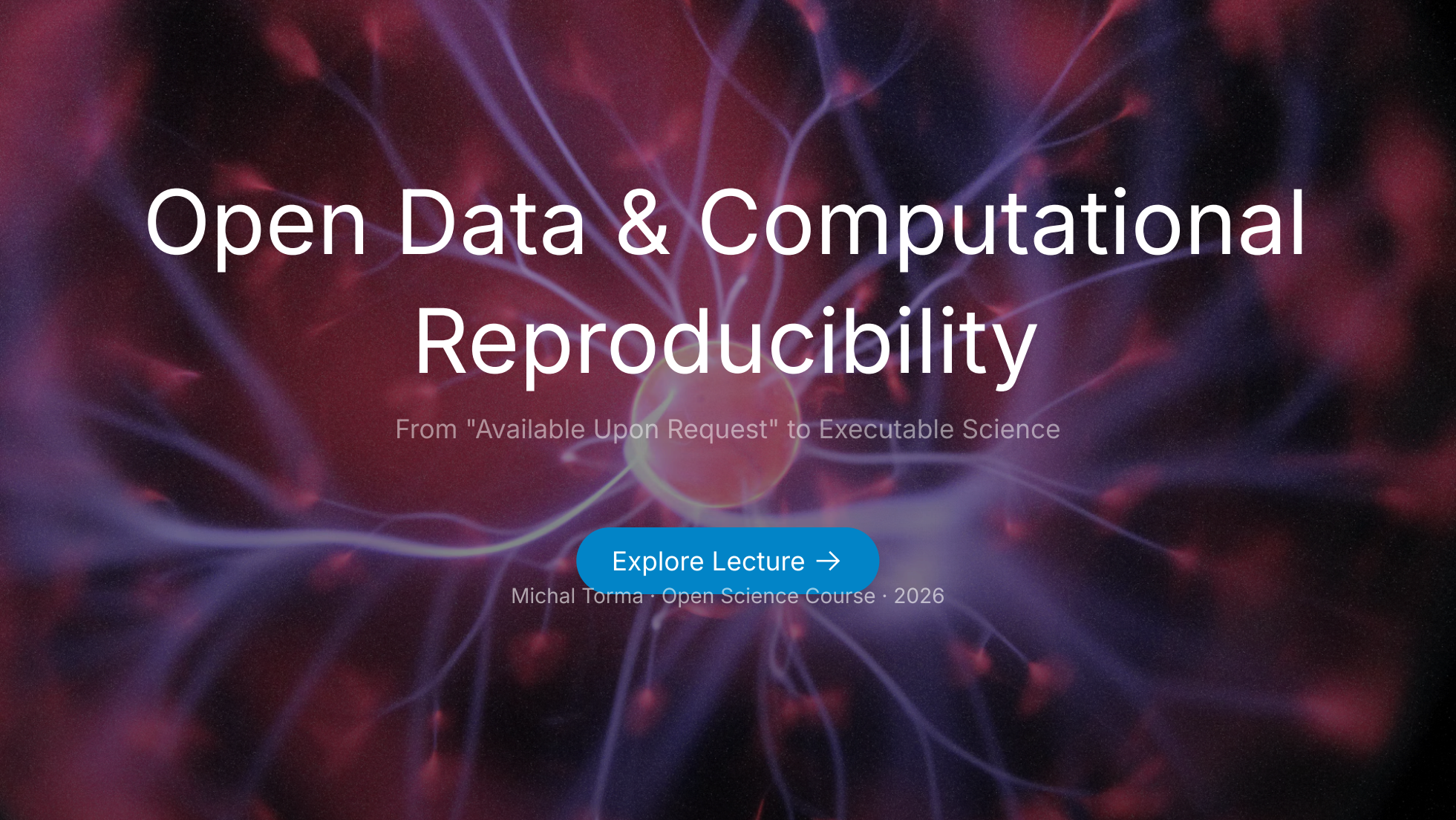

Open Data & Computational Reproducibility
From "Available Upon Request" to Executable Science
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Michal Torma · Open Science Course · 2026

The Computational Reproducibility Stack

How do we ensure computational results run 10 years from now?



1. Data & Metadata

- • Raw untransformed data
- • Tidy tabular formats (CSV, Parquet)
- • Data dictionary & schema
- • Zenodo / Dryad persistent DOIs



2. Code & Automation

- • Git version control
- • Modular analysis scripts
- • Literate notebooks (Jupyter/Quarto)
- • Pipeline engines (Nextflow, Snakemake)



3. Environment

- • Lockfiles (pip freeze, conda-lock)
- • OCI / Docker containers
- • Binderhub / Singularity execution
- • Hardware & OS documentation

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Reproducibility Checklist

- ✅ Code in public git repository with tag matching paper
- ✅ Data archived with immutable DOI on Zenodo
- ✅ Explicit open licenses for both code and data
- ✅ Environment specification (Dockerfile / conda-lock)
- ✅ One-click reproducibility script (e.g. `make run`)